Tianhao Li

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Executive Summary

Researcher focused on LLM agents, with particular interest in autonomy, practical application, and security in high-stakes domains. I aim to understand how autonomous agentic systems behave under real-world constraints, how they fail, and how they can be evaluated and aligned for safe, reliable use. My work emphasizes building rigorous yet practical methods that connect frontier AI research with applied security, trustworthy system design, and domain-specific risk assessment. My research and engineering capabilities are reflected in a strong track record of top-tier publications, and industry experience, thinking like a researcher while shipping like an engineer.

Education

- **Duke University** *2024-2026*
  Master of Science, Medical Physics (Computer Science Track, by Thesis) GPA: 3.5
- **North China University of Technology** *2020-2024*
  Bachelor of Engineering, Information Security GPA: 3.9

Experience

- **ByteDance** *2025/11-Present*
  Research Intern, Agent $\xrightarrow{FTE}$ *Evaluation Expert, Agent* Beijing, China
  - Developed a theoretical framework for agent trajectory evaluation and alignment, established annotation guidelines, constructed a trajectory-level failure attribution dataset, and designed evaluation algorithms for diagnosing agent trajectory failures, supporting post-training data generation for agent RL.
- **TOPSEC** *2024/03-2024/09*
  Security Researcher Intern Beijing, China
  - Built an end-to-end adversarial robustness evaluation prototype that turned scattered attack scripts into a systematic model stress-testing pipeline: generated perturbations under configurable threat models, executed batch robustness tests across ML classifiers, analyzed accuracy degradation and failure patterns under adversarial inputs, and surfaced model vulnerabilities through quantitative robustness metrics and reproducible evaluation reports.
- **NSFOCUS** *2023/09-2024/03*
  Security Researcher Intern Beijing, China
  - Built the security intelligence loop behind NSFOCUS LLM Security Assessment System (LSAS): translated trustworthy-AI research and Gartner threat reports into a structured LLM risk taxonomy, designed adversarial evaluation algorithms and prototype workflows, and validated them through offensive testing on public and private LLMs, ultimately shaping the system into a Red-Teaming-as-a-Service platform for scalable LLM vulnerability assessment.

Honors and Awards

- **Evaluation and Model Insight Award** EACL *2026*
- **Black Hat Asia Scholarship** Black Hat *2026*
- **AISST Technical Fellowship** Harvard AI Safety Student Team *2026*
- **Tinker Research Grant** Thinking Machines Lab *2025*
- **Outstanding Thesis Award** Undergrad *2024*
- **Outstanding Student Scholarship** Undergrad *2023*

Publications

A2ASecBench: A Protocol-Aware Security Benchmark for Agent-to-Agent Multi-Agent Systems *ICLR 2026*
Tianhao Li, Chuangxin Chu, Yujia Zheng, Bohan Zhang, Neil Zhenqiang Gong, Chaowei Xiao
International Conference on Learning Representations [paper]

Are All Prompt Components Value-Neutral? Understanding the Heterogeneous Adversarial Robustness of Dissected Prompt in Large Language Models *EACL 2026*
Yujia Zheng*, Tianhao Li*, Haotian Huang, Tianyu Zeng, Jingyu Lu, Chuangxin Chu, Yuekai Huang, Ziyou Jiang, Qian Xiong, Yuyao Ge, Mingyang Li
European Chapter of the Association for Computational Linguistics [paper]

SciSafeEval: A Comprehensive Benchmark for Safety Alignment of Large Language Models in Scientific Tasks *AICS 2025*
Tianhao Li, Jingyu Lu, Chuangxin Chu, Tianyu Zeng, Yujia Zheng, Mei Li, Haotian Huang, Bin Wu, Zuoxian Liu, Kai Ma, Xuejing Yuan, Xingkai Wang, Keyan Ding, Huajun Chen, and Qiang Zhang.
AAAI-25 Workshop on Artificial Intelligence for Cyber Security [paper]

A Survey of Trajectory Evaluation and Alignment for LLM Agents *Working Paper*
Tianhao Li, Jie Liu, Wenbo Shi, Yiyun Lu, Leran Chen, Yongshuai Li
Working paper

Retrogradus: Addressing Intent Deviation in Tool-Using Agents via Reverse Data Synthesis and Mutation on Critical Parameters *ArXiv 2026*
Qian Xiong, Yuekai Huang, Yujia Zheng, Tianhao Li, Ziyou Jiang, Zhiyuan Chang, Zhaoyang Li, Huanxiang Feng, and Mingyang Li
ArXiv [paper]

Butterfly Effects in Toolchains: A Comprehensive Analysis of Failed Parameter Filling in LLM Tool-Agent Systems *EMNLP 2025 Findings*
Qian Xiong, Yuekai Huang, Ziyou Jiang, Zhiyuan Chang, Yujia Zheng, Tianhao Li, and Mingyang Li
Findings of the Association for Computational Linguistics [paper]

Beyond Accuracy: Comprehensive Alignment for AI-Driven Molecular Design *AI4Sci 2026*
Qiang Zhang, Xiang Zhuang, Chenyi Zhou, Yihang Zhu, Tong Xu, Tianhao Li, Dianbo Liu, Shengchao Liu, Keyan Ding, Emine Yilmaz, Haofen Wang, and Huajun Chen
AI for Science [paper]

A Survey of Vibe Coding with Large Language Models *ArXiv 2025*
Yuyao Ge, Lingrui Mei, Zenghao Duan, Tianhao Li, Yujia Zheng, Yiwei Wang, Lexin Wang, Jiayu Yao, Tianyu Liu, Yujun Cai, Baolong Bi, Fangda Guo, Jiafeng Guo, Shenghua Liu, and Xueqi Cheng
ArXiv 2025 [paper]

From Droplets to Diagnosis: AI-Driven Imaging and System Integration in Digital Nucleic Acid Amplification Testing *Biosens. Bioelectron. 2025*
Yuanyuan Wei, Xianxian Liu, Yao Mu, Changran Xu, Guoxun Zhang, Tianhao Li, Zida Li, Wu Yuan, Ho-Pui Ho, and Mingkun Xu
Biosensors and Bioelectronics 2025 [paper]

AI Benchmarks Carpentry and Democratization *Under Review*
Gregor von Laszewski, Wesley Brewer, Jeyan Thiyagalingam, Juri Papay, Armstrong Foundjem, Piotr Luszczek, Murali Emani, Shirley V. Moore, Vijay Janapa Reddi, Matthew D. Sinclair, Sebastian Lobentanzer, Sujata Goswami, Benjamin Hawks, Marco Colombo, Nhan Tran, Christine R. Kirkpatrick, Abdulkareem Alsudais, Gregg Barrett, Tianhao Li, Kirsten Morehouse, Shivaram Venkataraman, Rutwik Jain, Kartik Mathur, Victor Lu, Tejinder Singh, Khojasteh Z. Mirza, Kongtao Chen, Sasidhar Kunapuli, Gavin Farrell, Renato Umeton, and Geoffrey C. Fox
Frontiers in High Performance Computing, Section Benchmarking [paper]

SFSWTS: A Spatial-Frequency Shifted Windows and Time Self-Attention Network for EEG Emotion Recognition *Neurocomputing 2025*
Weizhi Ma, Ying Li, Tianhao Li, Haowei Yang, Zhengping Li, Lijun Wang, and Junyu Xuan
Neurocomputing [paper]

Technical Reports

A Robust, Defensible, and Reproducible Methodology for Benchmarking Single-Turn Jailbreak Attacks on Large Language Models *2026*
MLCommons – mlcommons.org/2026/02/jailbreak-0-7 [paper]

Agentic Product Maturity Ladder V0.1 *2025*
MLCommons – mlcommons.org/ailuminate/agentic [paper]

AILuminate Security: Introducing v0.5 of the Jailbreak Benchmark from MLCommons *2025*
MLCommons – mlcommons.org/ailuminate/jailbreak [paper]

AILuminate: Introducing v1.0 of the AI Risk and Reliability Benchmark from MLCommons *2025*
MLCommons – mlcommons.org/en/ailuminate [paper]

Professional Services

- **Journal Referee:** ACM TIST, IEEE TAI, IEEE TBME, IEEE JBHI, IJHCI, ESWA, EAAI, RESS, Computer Networks, Pattern Recognition Letters, Pattern Recognition
- **Program Committee Member:** ICLR, NeurIPS, AAAI, ACL ARR, IJCAI, COLING, IJCNN, ISBI, CHIL, LLMSEC@ACL

Invited Talks

- From Language Models to Autonomous Agent Systems: An Introduction (20 mins). *MedPhy 751K, Duke Kunshan University*, Apr 2026, Jiangsu, China.
- Toward Trustworthy Generative Foundation Models and Autonomous Agent System (25 mins). *Institute of Software, Chinese Academy of Sciences*, Aug 2025, Beijing, China.

Open-Source Activity

- **NVIDIA/garak**
  github.com/nvidia/garak – Generative AI red-teaming & assessment kit Contributor
  - Extended the toolkit from text-only red-teaming to multimodal any-to-any assessment with #587: implemented a FigStep-based image+text-to-text jailbreaking pipeline for LLaVA.
  - Improved system reliability through bug fixes #296, #401, and enhanced exception handling #566 across core modules.